Fiscal Year:	FY 2007	**Task Last Updated:**	FY 03/09/2007
PI Name:	Morgan, William F. Ph.D.		
Project Title:	Mitigating High Z Radiation Induced Genomic Instability by Non-Protein Thiols		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	wfmorgan@som.umaryland.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	410 706-2475
Organization Name:	University of Maryland, Baltimore		
PI Address 1:	Radiation Oncology, 655 West Baltimore Street		
PI Address 2:	BRB 7-011		
PI Web Page:			
City:	Baltimore	**State:**	MD
Zip Code:	21201-1509	**Congressional District:**	7
Comments:	Deceased--PI passed away November 2015		
Project Type:	Ground	**Solicitation / Funding Source:**	2004 Radiation Biology NNH04ZUU005N
Start Date:	08/12/2005	**End Date:**	07/09/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: PI changed to Janet Baulch in 2008; project continues under Baulch; end date changed to 07/09/2008 (12/2010) NOTE: End date changed back to 5/31/2009 per S. Krenek/JSC (8/07) End date change to 9/30/2009 per JSC update info (10/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Baulch, Janet (University of Maryland)		
Grant/Contract No.:	NNJ06HD31G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	That astronauts will be exposed to the space radiation environment is unavoidable, and all radiation exposure has the potential to elicit a carcinogenic response. Cancer is characterized by gross destabilization of the genome. In vivo and in vitro exposure to radiation can lead to genomic instability. This radiation induced genomic instability shares many of the characteristics associated with genetic instabilities observed in human tumors and may well be process that transforms a normal cell to a neoplastic growth following radiation exposure. We have developed a rapid and sensitive assay for measuring induced genomic instability occurring in the progeny of irradiated cells. One way to protect space travelers from the deleterious effects of radiation exposure is with the use of radioprotective compounds. In this application we will investigate the radioprotective effect of three non-protein thiol drugs already approved for clinical use. This will be achieved by investigating three specific aims: Specific aim 1 will test the hypothesis that non-protein thiols can reduce the frequency of genomic instability induced by High Z particles in human cells. These human cells have been transfected with a sensitive and reliable reporter system to analyze delayed recombination/deletion events associated with radiation exposure in the progeny of surviving cells. Specific aim 2 will test the hypothesis that some of the delayed effects of high Z radiation result from nontargeted bystander effects resulting from soluble factors secreted by irradiated cells. Specific aim 3 will investigate the molecular mechanism(s) by which the non-protein thiols exert their protective effect. Specifically we will test the hypothesis that treating cells with these drugs induces elevated intracellular levels of manganese superoxide dismutase and this reduces iron ion induced genomic instability in RKO36 cells. The goal of these studies is to determine whether drugs already approved for use in the human population can reduce genomic instability induced by radiations commonly encountered in space. If so, then these drugs can potentially be taken by astronauts to protect against deleterious effects associated with space radiations.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	These studies will test the hypothesis that non-protein thiols (NPTs) can protect astronauts from the detrimental effects of space radiation exposure and will evaluate cells surviving exposure to high Z particles to determine whether they are at increased risk for genomic instability. These studies will also evaluate the molecular mechanisms by which the NPTs exert their protective effect. The clinic can utilize a greater understanding of how these NPTs protect against ionizing radiation induced effects to improve radiotherapy treatments. Additionally, demonstrating that administration of an NPT to astronauts immediately after irradiation might reduce the detrimental effects would also have application for homeland security in the event of a radiological terrorist attack.
Task Progress:	Task Progress: Specific Aim 1 tests the hypothesis that non-protein thiols can reduce the frequency of genomic instability in human cells as measured by delayed recombination/deletion events using a GFP reporter assay. Specific Aim 2 tests the hypothesis that some of the effects of high Z radiation stem from non-targeted bystander effects. Specific Aim 3 evaluates the molecular mechanisms by which non-protein thiols exert their protective effects. During year 2 we have made significant progress on this project. We participated in our first beam run during September of 2006, NSRL06C and we are currently preparing for our second run at BNL during NSRL07A. Analyses of data and of RKO36 clones from the first run yielded promising preliminary results. These preliminary results suggest that; 1) both high LET Fe-ion radiation and low LET x-rays induce delayed recombination and/or mutations/deletions in the RKO36 green fluorescent protein (GFP) experimental system, 2) WR-1065 protects RKO36 cells from radiation-induced delayed recombination and/or mutations/deletions, and 3) WR-1065 protects RKO36 cells from radiation-induced chromosomal damage, hypoxanthine-guanine phosphoribosyltransferase (HPRT) mutations, increases in reactive oxygen species (ROS) levels and cell death. We will build upon these preliminary results during the Spring 2007 beam run and have applied for additional beam time during 2007-2008. Additionally, two of our graduate students have applied to attend NASA's 2007 Space Radiation Summer School.
Bibliography Type:	Description: (Last Updated: 02/02/2009)
Abstracts for Journals and Proceedings	Dziegielewski J, Baulch JE, Goetz W, Murley JS, Grdina DJ, Morgan WF. "Amifostine metabolite WR-1065 protects irradiated cells from genetic instability." Space Radiation (Poster Session 2). NASA Human Research Program Investigators' Workshop, League City, Texas, February 12-14, 2007. NASA Human Research Program Investigators' Workshop, February 2007. February 12-14, 2007 League City, Texas , Feb-2007
Articles in Peer-reviewed Journals	Murley JS, Kataoka Y, Baker KL, Diamond A, Morgan WF, Grdina DJ. "Manganese superoxide dismutase (SOD2) mediated delayed radioprotection induced by the free thiol form of Amifostine and tumor necrosis factor-alpha." Radiation Research, in press, February 2007. , Feb-2007